

Malware Math

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1 Introduction

The claim mind model, while a decent conceptual analog, still retained a few key flaws. Defense collapses to pruning for minimal graphs, offense collapses to asserting end goal states without nuance, malware representations could not operate autonomously or replicate, and capabilities were external to the system. This note introduces a system which is closed under its own semantics, but may optionally introduce side effects based on the implementation of the language.

2 Formulation

A term is a rewrite rule. On the left of a term are the conditions for the rewrite, on the right is the resulting rewrite. Programs consist of a series of terms, the act of writing a program and the act of evaluating it are the same action, as atoms are appended to the program concatenatively, triggering all previous rules to run in the process. Rewrites occur from left to right. Signals in this model consist of atoms being appended to the end of the stream, triggering autometabolism. Capabilities occur naturally as computed effects the system is capable of. Side effects are necessary for capabilities which effect the environment the program exists in.

3 Immunity

Defense in this model still involves preserving capabilities, doing so though writing rules which prevent harmful capabilities from being generated as rules, and disallowing potentially harmful patterns from introducing malware.

4 Malware and Virality

Malware occurs when a rewrite rule is successfully generated adversarially in a program stream against the wishes of the immune system. Malware in this model can finally be truly viral and self replicating, and can even attack the immune system.